# Group 9 — Presentation script (≈ 6 minutes)

Two presenters: **A** and **B**. Read aloud at a normal pace (~150 words/minute). Lines in *italics* are stage directions, not spoken. Time budget per slide is shown in **[brackets]**.

The full delivery comes to **about 6 minutes**, leaving a 1-minute buffer for a slow opening or extra Q&A.

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## Slide 1 — Title [15 s · A]

Hi, we're Group 9. Today we'll show how we built a system that watches a video, picks out the moments that matter, and stitches them into a short summary clip — the way a movie trailer summarises a film.

*(A nods to B.)*

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## Slide 2 — Four very different videos [30 s · A]

We worked with four short videos from the TVSUM dataset. A graduation flash mob with a long speech. A street flash mob in India during a cricket world cup. A flash-mob orchestra on a metro train. And a beekeeper interview with a long voice-over. We picked these on purpose — they look very different from each other. That way we can see where our system holds up and where it falls apart.

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## Slide 3 — Six steps, one notebook [35 s · A]

Our pipeline has six steps. *(point at each box as you go)* One: **annotate**. We mark the important moments by hand. Two: **agreement**. We measure how much the two of us agreed. Three: a **vision-language model** lists what's happening in the video. Four: a **search model** uses those descriptions to find the exact start and end times. Five: we **score** how close the predictions are to our annotations. And six: we **stitch the predicted clips** into one summary video. All of this lives in a single notebook plus a small Python package. Over to you, **B**.

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## Slide 4 — Two annotators rarely fully agree [50 s · B]

Before trusting any model, we wanted to know how much *we* agreed with each other. We measured three things. First, **did we mark the same moments as eventful?** Numbers ranged from 0.15 to 0.48 — moderate. We picked broadly the same parts of each video, but we disagreed on how many separate events to draw. Second, **when we did agree on an event, did we draw the same start and end times?** Here we did much better — between 0.55 and 0.72 overlap. So we tend to draw similar windows, we just don't always agree on how to slice them up. Third, **did we agree on which events were essential?** This is where it falls apart. The beekeeper video is below chance — meaning we *disagreed more than random.* One of us prioritised the spoken explanations, the other prioritised the visual demonstrations. Both are valid, and that's the whole point of subjectivity.

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## Slide 5 — Three ways of asking the same question [50 s · B]

For Step 3 we used a small vision-language model called **SmolVLM2** — basically an AI that reads video frames and writes about them. We tried three different ways of asking it the same question. *(point to first card)* **"Just give me the events"** — works best. Four to five short events per video, parsed cleanly. *(point to second card)* **"Events with timestamps"** — almost always fails. The model isn't trained to know *when* things happen, so it either refuses or returns one window covering everything. *(point to third card)* **"Caption every frame, then merge"** — hit-or-miss. Works on the beekeeper video, where it covers 42 % of our annotated events; doesn't work elsewhere. The big failure across all three: **anything that depends on speech is invisible to a vision model.** A spoken joke turns into "a man stands at a microphone".

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## Slide 6 — A bigger model is the cheapest improvement [55 s · A]

Now Step 4 — finding *when* each event happens. We tried two versions of the same approach, using a model called CLIP that matches text to images. *(point to chart)* The smaller, faster CLIP got an overlap score of 0.09. The larger, slower CLIP got 0.11 — a 27 % improvement just from a bigger backbone. The bigger one also finds the right rough region 21 % of the time, vs 14 % for the smaller one. We also tried something the assignment suggests: rephrasing each query three different ways and combining the results. Surprisingly, **it didn't help** — and sometimes made things slightly worse. We think averaging three rephrased queries blurs the strongest signal in the original. Breaking it down by what kind of event: performance scenes get 0.17 overlap, the beekeeper gets 0.07, and speech is **zero**.

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## Slide 7 — Walk-through: the train orchestra [75 s · B]

Let's walk through one full example. This is video 35 — the classical-music flash mob on a metro train. *(point to left column)* We annotated 13 key moments — from the announcement on the screen, through each musician revealing themselves, to the climax with the full orchestra. *(point to right column)* Our AI listed four salient events: a flutist, a group of violins, passengers reacting, and the train arriving. Our search model — CLIP-Large — then located those four events. On this video it scored its best result of all four: **0.33 mean overlap, finding the right region 75 % of the time.** Let's watch the summary it produced. *(play* `outputs/summaries/video_35_summary.mp4`*, 28 seconds, 8 clips)* What it gets right: the orchestra reveal, the violins joining in, the climax. What it misses: the conductor — that's a single person standing up briefly — and the closing text overlay that reveals it's an advertisement, because the model can't read text.

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## Slide 8 — Three things we learned [40 s · A]

Three takeaways. One. **Subjectivity matters more than the choice of model.** Two humans annotating the same video already disagree on what matters. An AI working from one of those views will reflect *that* bias. Two. **Off-the-shelf vision models work well on what they can see, and fail on what they can only hear.** Performances and dance: good. Speeches and jokes: zero overlap. There's no shortcut around this — we need speech recognition. Three. **A bigger model is the cheapest single improvement.** Just swapping the small CLIP for a larger one gave us a 27 % gain with no other changes.

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## Slide 9 — What we'd do with another week [25 s · B]

If we had more time, we'd do three things. One: add **speech recognition**, so dialogue events stop being invisible. This is the single biggest win available. Two: re-rank clips by **importance**, not just relevance, so the summary respects our saliency ratings. Three: bring back **CG-DETR**, the model the assignment originally suggested. Its checkpoint is currently offline, but if it returns, that's our second-method comparison done properly.

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## Slide 10 — Thank you [5 s · A]

Thank you. Happy to take questions.

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## Word count and timing

| Slide | Words spoken | Target time |
|---|---|---|
| 1 | 35 | 0:15 |
| 2 | 70 | 0:30 |
| 3 | 95 | 0:35 |
| 4 | 145 | 0:50 |
| 5 | 145 | 0:50 |
| 6 | 145 | 0:55 |
| 7 | 175 | 1:15 (incl. clip) |
| 8 | 105 | 0:40 |
| 9 | 65 | 0:25 |
| 10 | 7 | 0:05 |
| **Total** | **~990** | **~6:00** |

If you want to land at 7 minutes, slow down the delivery on slides 4, 5, 6 and pause for 2–3 seconds between bullets — these are where the audience needs the most processing time. If you want to land at 5 minutes, cut the second example in slide 5 ("doesn't work elsewhere") and trim slide 9 to one bullet.

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## Q&A cheat-sheet

**Why CLIP and not CG-DETR (the demo's recommended model)?**

> "We tried — its official checkpoint isn't downloadable anymore. The lighthouse repo's download script returns 404 and the author's HuggingFace account doesn't host it. We used two CLIP backbones to still have a model-size comparison."

**Why is your overlap so low (~0.11)?**

> "Zero-shot CLIP has no temporal training. Published CG-DETR trained on QVHighlights gets ~0.40 — about 4× better — because it learns the localisation task end-to-end."

**Why does paraphrase fusion *hurt*?**

> "Our hypothesis: averaging text embeddings of three rephrasings blurs the most discriminative direction in embedding space. CG-DETR-style models that handle multiple queries internally probably profit from the same input."

**How did you choose the 1.5 s minimum clip length?**

> "Empirical — clips shorter than that never contained enough of the event to be useful, and they made the summaries feel jittery. Anything between 1 s and 2 s worked similarly; we settled on 1.5 s."

**How did you handle the long graduation video (video 33, 7 min)?**

"CLIP is per-frame, so video length doesn't matter for our retrieval method. We sample at 1 Hz regardless of duration."

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## Glossary (for if the audience asks)

- **Overlap / IoU**: how well a predicted time-window matches a true one. 1.0 is identical, 0.0 means no overlap.
- **Cohen's Kappa (κ)**: agreement between two raters, corrected for chance. 1.0 = perfect, 0.0 = random, negative = worse than random.
- **CLIP**: a model that learns to match text descriptions to images.
- **VLM**: a vision-language model — takes images and text, produces text.
- **Zero-shot**: using a pretrained model on a new task without retraining it.